



June 17, 2026

Dear Dr. Varga,

I am writing to express my enthusiastic interest in the PhD position at Loughborough University's Artificial Intelligence for the Built Environment (AI4BE) program. As an AI Data Science Engineer with a passion for interdisciplinary research, I am excited about the opportunity to apply my skills in AI, architecture, and business to real-world problems.

My background is rooted in engineering, with a strong foundation in computer science and mathematics. My recent project, Cardiovascular Medical Digital Twin, has allowed me to explore the intersection of AI and medicine. Specifically, my work on *MedicalMambaTS, a linear – time state space model for sub – millisecond bedside inference, demonstrates my ability to develop efficient AI solutions that can be applied to various domains.*

I am particularly drawn to this PhD opportunity because it aligns with my interests in applying AI to improve the built environment. The project's focus on using AI to contribute to the design, build, operate, and decommission of buildings resonates with me. My experience working with PyTorch Autograd to develop a 0D Windkessel Physics-Informed Operator Network (PI-DeepONet) has given me a solid understanding of the importance of physical constraints in AI applications.

I believe that my unique blend of technical skills and passion for interdisciplinary research makes me an ideal candidate for this PhD program. I am confident that my expertise will enable me to make significant contributions to the AI4BE project, and I look forward to the opportunity to discuss my application with you further.

Thank you for considering my application. I have attached my CV and a summary of my project achievements.

Sincerely,

Ismail Aissaoui